Docker Compose Operations - nap-dev-fullstack

All commands assume you are in the project root where `docker-compose.yml` lives:

```
cd /home/zulu/Projects/containers/nap-dev-fullstack
```

1. Start all service containers (detached mode)

```
docker compose up -d
```

- **-d** / **--detach** runs the containers in the background and returns control to your shell.
- Compose enforces the start order automatically: **ict-db** starts first, **ict-api** waits until **ict-db** is **healthy**, then **ict-web** waits until **ict-api** is **healthy**.
- If images are missing locally, they are pulled first (note: **ict-db** pulls from **mcr.microsoft.com**, not Docker Hub).

Useful variations:

```
# Recreate containers even if config/images unchanged
docker compose up -d --force-recreate

# Pull newer images before starting
docker compose up -d --pull always
```

2. Check the status of all containers

```
docker container ls
```

Shows each running container with its ID, image, command, state, and published ports. Example output:

```
CONTAINER ID   IMAGE                                        COMMAND                  CREATED      STATUS                PORTS                                                NAMES
18462fc091c2   debian:trixie                                "sleep infinity"         2 days ago   Up 2 days             0.0.0.0:5500->5500/tcp, [::]:5500->5500/tcp          ict-web
f0d0c69e824a   node:26-trixie                               "docker-entrypoint.s…"   2 days ago   Up 2 days (healthy)   0.0.0.0:3080->3080/tcp, [::]:3080->3080/tcp          ict-api
fb50402baccc   mcr.microsoft.com/mssql/server:2022-latest   "/opt/mssql/bin/laun…"   2 days ago   Up 2 days (healthy)   0.0.0.0:1433->1433/tcp, [::]:1433->1433/tcp          ict-db
```

- The **(healthy)** tag in the **STATUS** column reflects the configured healthchecks (**ict-db** runs a **sqlcmd SELECT 1** probe; **ict-api** verifies the Node runtime and a writable **/workspace**).
- **ict-web** has no healthcheck, so it just shows **Up**.

Useful variations:

```
# Include stopped/exited containers
docker container ls -a

# Follow logs to diagnose a container
docker logs -f ict-db

# Inspect one container in detail
docker inspect ict-api
```

3. Shut down all service containers

```
docker compose down
```

- Stops and **removes** the containers and the Compose-created network.
- Your bind-mounted data in **workspace/db** (SQL Server files) and **workspace/web** / **workspace/api** is **not** deleted - it lives on the host.

Useful variations:

```
# Stop containers but keep them (start again with `docker compose start`)
docker compose stop

# Start them again after `stop`
docker compose start

# Shut down AND remove named volumes (not needed here - bind mounts only)
docker compose down -v
```

Quick reference

Goal	Command
Start everything, background	`docker compose up -d`
See status	`docker container ls`
Stop & remove	`docker compose down`